Science Advances
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Supplementary Materials for

Global agreement masks regional divergence in soil respiration

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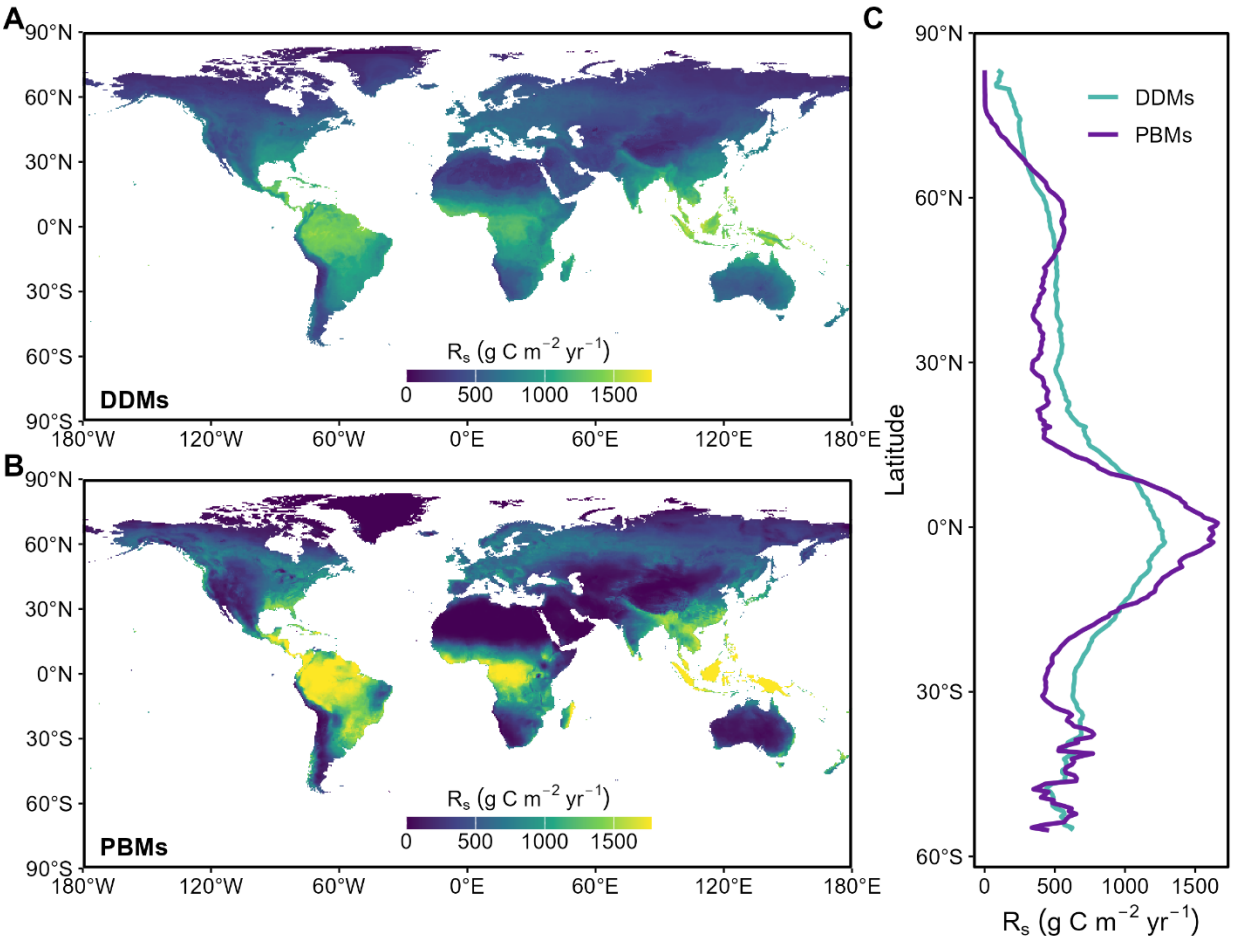


Fig. S1.

Difference in the spatial distribution of global soil respiration (Rs) is estimated by data-driven models (DDMs) and process-based models (PBMs).

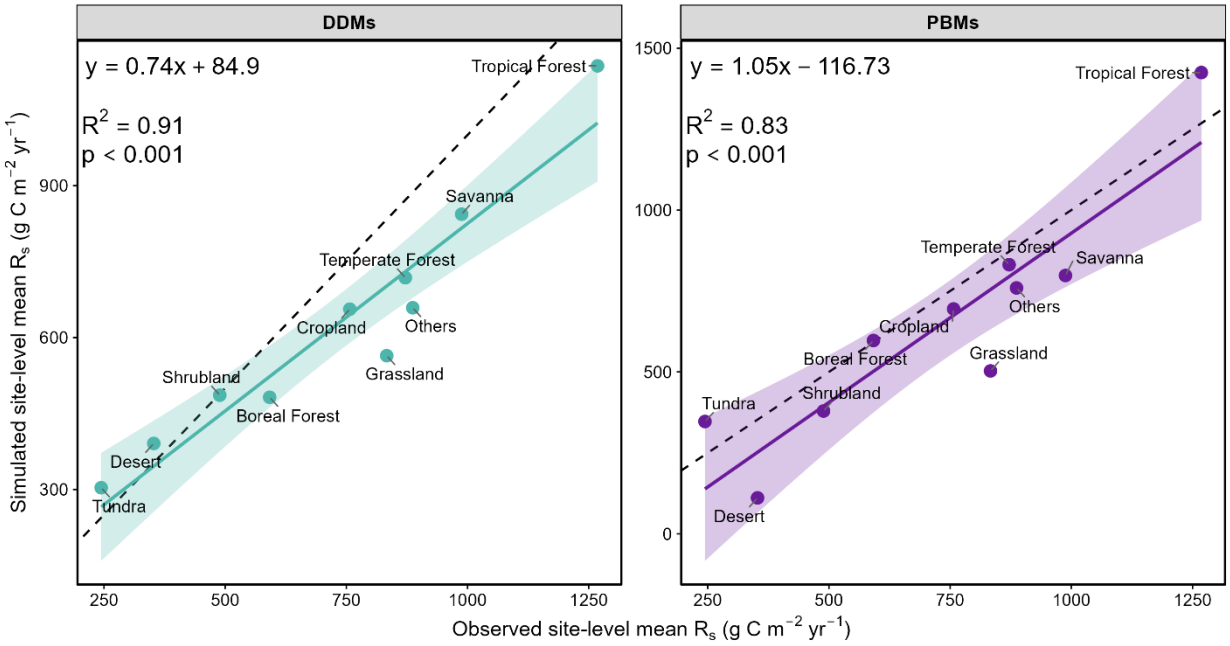


Fig. S2.

Comparison of biome-level mean soil respiration (Rs) between SRDB-V5 and model estimates. Model outputs from data-driven models (DDMs) and process-based models (PBMs) were extracted at the SRDB-V5 site locations and averaged across sampling sites within each biome.

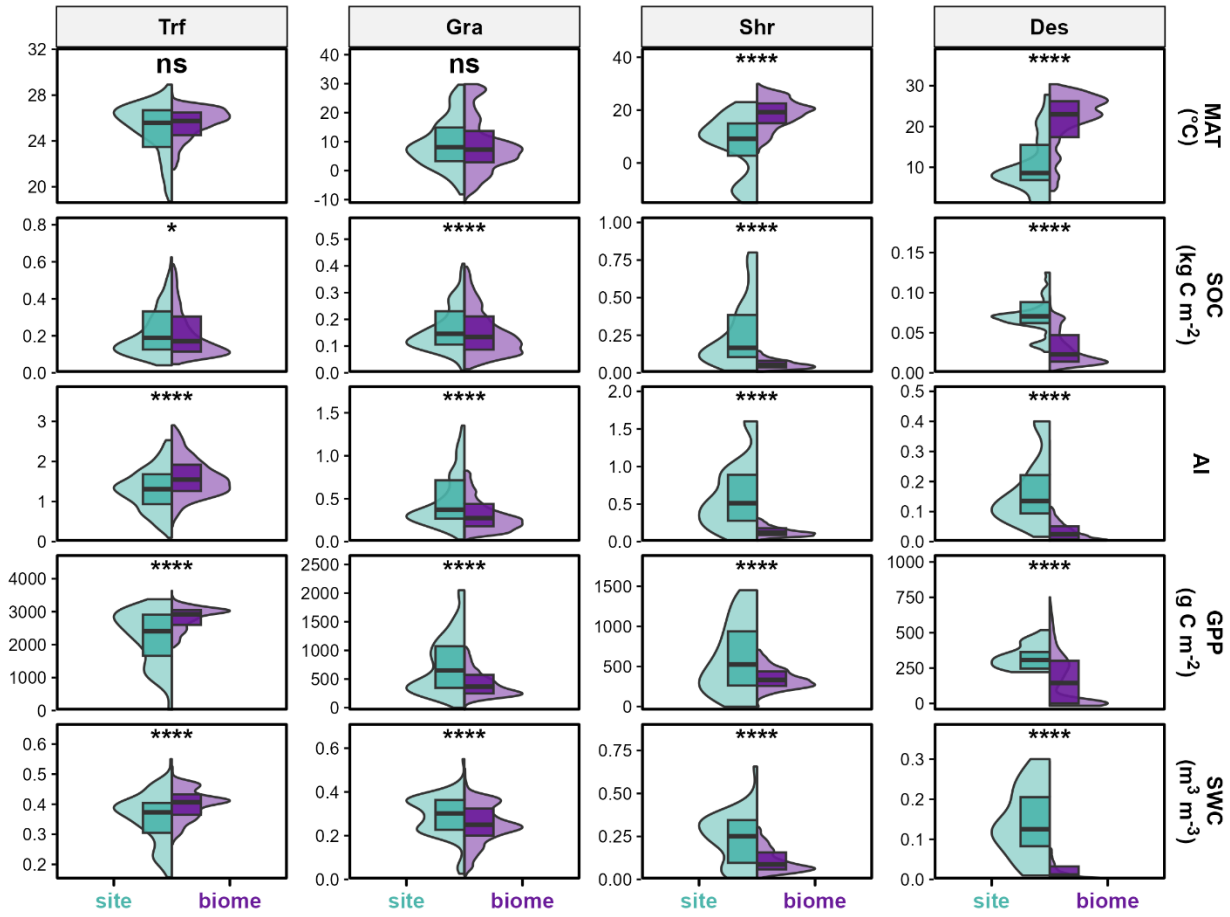


Fig. S3.

Representativeness of observation sites relative to biome-wide conditions for environmental drivers and modelled soil respiration (Rs) across four biomes. Split violin plots compare the distribution of each variable at SRDB-V5 measurement locations (site, left half, teal) with the distribution across all grid cells of the corresponding biome (biome, right half, purple). Half violins show kernel density estimates for site and biome distributions, truncated at the box-plot whisker ends. Boxes indicate the 25th, 50th (median) and 75th percentiles, with whiskers extending to the most extreme values within 1.5 × the interquartile range (IQR); values beyond the whiskers are not shown. Asterisks indicate significance from two-sided Wilcoxon rank-sum tests comparing site and biome distributions (****$p < 0.0001$, ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$; ns, $p > 0.05$). Biome abbreviations are defined in Fig. 1.

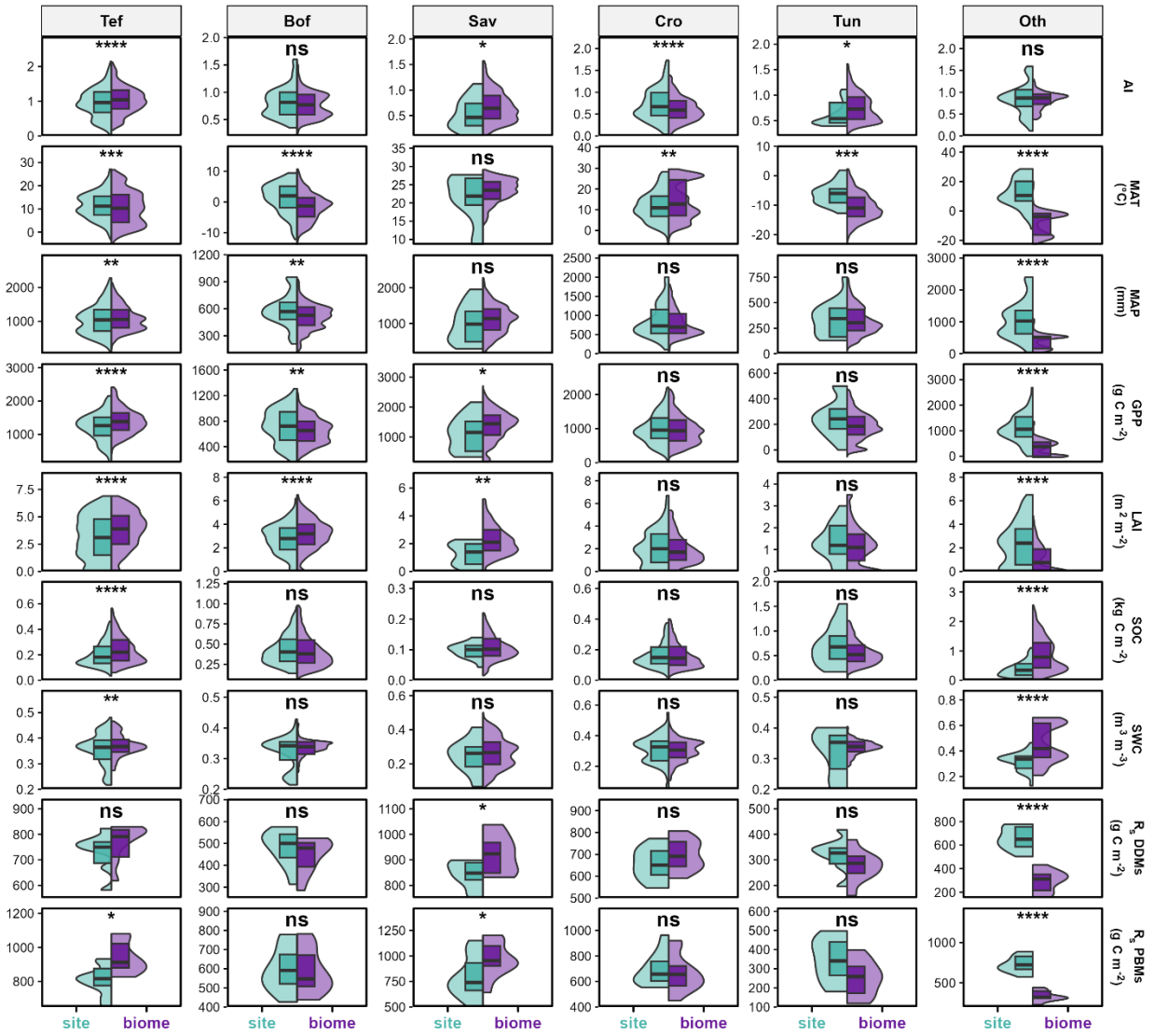


Fig. S4.

Representativeness of observation sites relative to biome-wide conditions for environmental drivers and modelled soil respiration (Rs) across four biomes. Split violin plots compare the distribution of each variable at SRDB-V5 measurement locations (site, left half, teal) with the distribution across all grid cells of the corresponding biome (biome, right half, purple). Half violins show kernel density estimates for site and biome distributions, truncated at the box-plot whisker ends. Boxes indicate the 25th, 50th (median) and 75th percentiles, with whiskers extending to the most extreme values within 1.5 × the interquartile range (IQR); values beyond the whiskers are not shown. Asterisks indicate significance from two-sided Wilcoxon rank-sum tests comparing site and biome distributions (****$p < 0.0001$, ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$; ns, $p > 0.05$). Biome abbreviations are defined in Fig. 1.

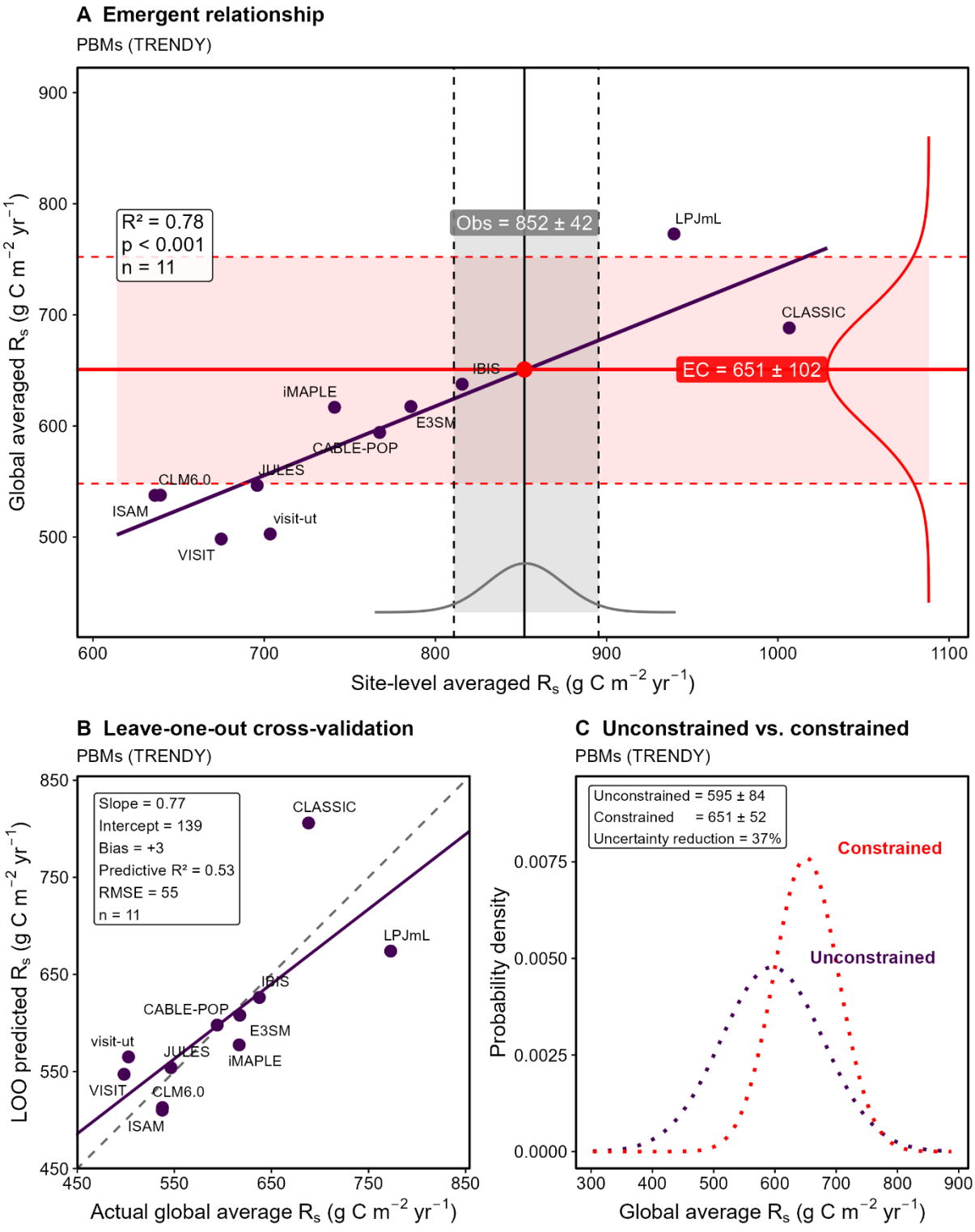


Fig. S5.

Emergent constraint on the global average soil respiration (Rs) of process-based models (PBMs). (A) Global average Rs versus site-level average Rs, the latter evaluated at the SRDB-

V5 observation locations, across the PBMs (points, n = 11). The solid line is the ordinary least squares regression across the models. The vertical dashed line marks the observational constraint given by the SRDB-V5 site-level mean. The red diamond and the horizontal solid line mark the constrained estimate of global average Rs obtained by mapping the observation through the regression, and the horizontal dashed lines mark its uncertainty range (mean ± 1 s.d.). The inset reports the coefficient of determination, the regression p value, and the number of models. (B) Probability density of global average Rs before (unconstrained) and after (constrained) the emergent constraint. The unconstrained density is defined by the mean and standard deviation across the model ensemble, and the constrained density was obtained by propagating the observational uncertainty and the regression uncertainty through the intermodel relationship in (C). The inset reports the mean ± 1 s.d. of each distribution and the relative reduction in the uncertainty range.

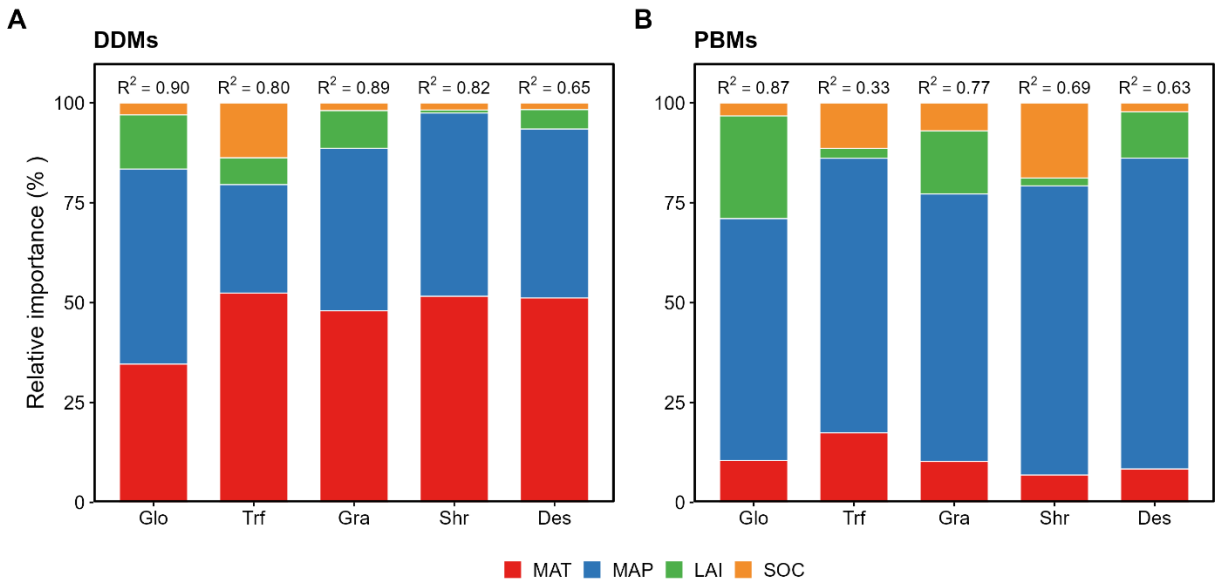


Fig. S6.

Relative importance of MAT, MAP, LAI and SOC in explaining soil respiration (Rs) in (A) data-driven models (DDMs) and (B) process-based models (PBMs). Importance is the absolute standardized ridge-regression coefficient of each predictor as a percentage of the sum across the four predictors (values sum to 100% per bar); R^2 above each bar is the variance in Rs explained by the model. Glo represents the global scale, and biome abbreviations are defined in Fig. 1.

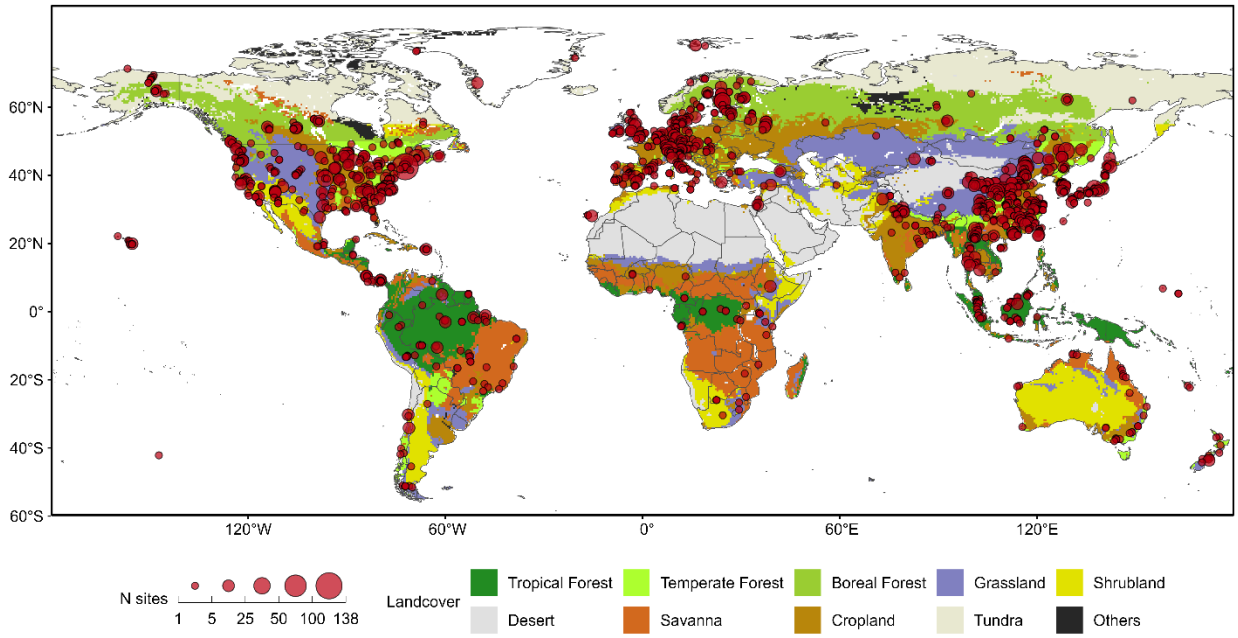


Fig. S7.

Global distribution of soil respiration (Rs) observations across biomes. Global distribution of Rs observation sites included in the analysis (red circles) overlaid on the spatial distribution of the ten biome categories (background colours). Circle size indicates the number of soil respiration observation sites within each grid cell (N sites).

Table S1.

Summary of historical global annual soil respiration (Rs) estimates based on data-driven models (DDMs). The 10 gridded Rs products used for this study analyses are highlighted in bold black.

ID	Study	Rs (Pg C /yr)	Method	Study Year	Temporal resolution	Spatial resolution	Drivers
1	Schlesinger (1997)(*75*)	75.0	Carbon balance		Annual		Climate
2	Raich and Schlesinger (1992)(*76*)	68.0	Area × Mean		Annual		Climate+Soil +Vegetation
3	**Raich and Potter (1995)**(*77*)	**76.5**	**Semi-empirical model**		**Monthly**	**0.5°**	**Climate**
4	Peng and Apps (2000)(*78*)	69.0	Semi-empirical model		Annual	0.5°	Climate
5	**Raich et al. (2002)**(*79*)	**80.4**	**Semi-empirical model**	**1980-1994**	**Monthly**	**0.5°**	**Climate**
6	Bond-Lamberty and Thomson (2010)(*80*)	98.0	Linear model	2008	Annual		Climate+Vegetation
7	**Hashimoto (2012)**(*81*)	**78.0**	**Semi-empirical model**	**1980-2009**	**Monthly**	**0.5°**	**Climate**
8	Chen et al. (2013)(*82*)	94.4	Semi-empirical model	1970-2008	Annual	0.5°	Climate+Soil
9	**Hashimoto et al. (2015)**(*12*)	**91.0**	**Semi-empirical model**	**1965-2012**	**Monthly**	**0.5°**	**Climate**

10	Oertel et al. (2016)(*83*)	95.0	Area × Mean		Annual		
11	Xu and Shang (2016)(*84*)	94.3	Area × Mean	2000–2012	Annual		
12	Adachi et al. (2017)(*85*)	94.8	Semi-empirical model	2001	Daily	4km	Climate+Soil
13	Adachi et al. (2017)(*85*)	93.8	Semi-empirical model	2009	Daily	4km	Climate
14	**Zhao et al. (2017)**(*13*)	**93.3**	**Machine learning**	**1960-2012**	**Annual**	**~9 km**	**Climate+Soil +Vegetation**
15	Hursh et al. (2017)(*35*)	93.9	Statistical model	2014	Annual		Climate+Soil +Vegetation
16	Hursh et al. (2017)(*35*)	80.3	Semi-empirical model	2014	Annual		Climate+Soil +Vegetation
17	Hursh et al. (2017)(*35*)	108.6	Semi-empirical model	2014	Annual		Climate+Soil +Vegetation
18	Jian et al. (2018)(*86*)	78.8	Machine learning	1961-2014	Monthly	0.5°	Climate+Vegetation
19	Jian et al. (2018)(*86*)	88.2	Machine learning	1961-2014	Annual	0.5°	Climate+Vegetation
20	Jian et al. (2018)(*86*)	70.9	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
21	Jian et al. (2018)(*86*)	66.6	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation

22	Jian et al. (2018)(*86*)	75.8	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
23	Jian et al. (2018)(*86*)	71.4	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
24	Jian et al. (2018)(*86*)	81.0	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
25	Jian et al. (2018)(*86*)	76.2	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
26	Jian et al. (2018)(*86*)	74.4	Semi-empirical model	1961-2014	Daily	0.5°	Climate+Vegetation
27	Jian et al. (2018)(*86*)	93.3	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
28	Jian et al. (2018)(*86*)	88.6	Semi-empirical model	1961-2014	Monthly	0.5°	Climate+Vegetation
29	Jian et al. (2018)(*86*)	100.7	Semi-empirical model	1961-2014	Annual	0.5°	Climate+Vegetation
30	Jian et al. (2018)(*86*)	97.0	Semi-empirical model	1961-2014	Annual	0.5°	Climate+Vegetation
31	Jian et al. (2018)(*66*)	78.3	single model	1961-2014	Monthly	0.5°	Climate
32	Jian et al. (2018)(*66*)	72.6	climate-specific model	1961-2014	Monthly	0.5°	Climate

33	**Warner et al. (2020)**(*23*)	**87.9**	**Machine learning**	**1961-2011**	**Annual**	**1km**	**Climate+Vegetation**
34	**Tang et al. (2019; 2020)**(*87*, *88*)	**101.0**	**Machine learning**	**1980-2012/ 2016**	**Annual**	**0.5°**	**Climate+Soil**
35	**Huang et al. (2020)**(*14*)	**72.6**	**Machine learning & Semi-empirical model**	**2000-2014**	**Annual**	**1km**	**Climate+Vegetation**
36	Lei et al. (2021)(*89*)	85.6	Linear model	1987–1998	Annual		Climate
37	Lei et al. (2021)(*89*)	87.5	Linear model	1999-2016	Annual		Climate
38	**Lu et al. (2021)**(*43*)	**85.5**	**Machine learning**	**1982–2012**	**Annual**	**0.5°**	**Climate+Vegetation**
29	**Stell et al. (2021)**(*18*)	**88.6**	**Machine learning**		**Annual**	**1km**	**Climate+Vegetation**
40	**Stell et al. (2021)**(*18*)	**96.5**	**Machine learning**		**Annual**	**1km**	**Climate+Vegetation**
41	Jian et al. (2022)(*8*)	68.0	Carbon balance		Annual	0.5°	
42	Jian et al. (2022)(*8*)	93.0	Machine learning		Monthly	0.5°	Climate+Vegetation
43	Jiang et al. (2024)(*65*)	95.1	Machine learning	2020	Monthly	0.5°	
44	Jiang et al. (2024)(*65*)	97.8	Machine learning	2020	Monthly	0.5°	
45	Jiang et al. (2024)(*65*)	85.9	Machine learning	2020	Monthly	0.5°	
43	Jiang et al. (2024)(*65*)	106.0	Machine learning	2020	Monthly	0.5°	Climate+Soil+Vegetation

47	Jiang et al. (2024)(*65*)	108.7	Machine learning	2020	Monthly	0.5°	Climate+Soil +Vegetation
48	Laffitte et al. (2025)(*46*)	79.4	Machine learning	1982-2018	Monthly	0.05°	Climate+Vegetation
49	Laffitte et al. (2025)(*46*)	78.3	Machine learning	1982-2018	Annual	0.05°	Climate+Vegetation
50	Jiang et al. (2025)(*24*)	85.1	Machine learning		Monthly	0.5°	Climate+Vegetation
51	Jiang et al. (2025)(*24*)	88.4	Machine learning		Monthly	0.5°	Climate+Vegetation

Table S2.
Detailed information of the TRENDY v13 Dynamic Global Vegetation Models used in this study.

Model	**Scenario**	**Spatial resolution**	**Reference**
CABLE-POP	S3	1°	Haverd et al. (2018)(*90*)
CLASSIC	S3	1°	Melton et al. (2020)(*91*)
CLM6.0	S3	~1.25°	Lawrence et al. (2019)(*92*)
E3SM	S3	~1.25°	Yang et al. (2023)(*93*)
IBIS	S3	0.5°	Xia et al. (2024)(*94*)
ISAM	S3	0.5°	Shu et al. (2020)(*95*)
JULES	S3	0.5°	Wiltshire et al. (2021)(*96*)
LPJml	S3	0.5°	Schaphoff et al. (2018)(*97*)
visit-ut	S3	0.5°	Ito (2019)(*98*)
VISIT	S3	0.5°	Kato et al. (2013)(*99*)
iMAPLE	S3	1°	Yue et al. (2024)(*100*)

Table S3.
Environmental variables used in this study and their data sources.

Variable	Description	Unit	Spatial resolution	Data source	Study
MAT	mean annual temperature	°C	0.5°	CRU TS v4.04	Harris et al. (2020)(*101*)
MAP	mean annual precipitation	mm yr^{-1}	0.5°	CRU TS v4.04	Harris et al. (2020)(*101*)
LAI	leaf area index	m^2 m^{-2}	1km	MODIS LAI	Yuan et al. (2011)(*56*)
SOC	soil organic carbon content	kg C m^{-2}	250 m	SoilGrids 2.0	Poggio et al. (2021)(*102*)
GPP	gross primary productivity	g C m^{-2} yr^{-1}	0.5° (RS+METEO)	FLUXCOM Carbon Fluxes (Annual Ensemble)	Tramontana et al. (2016) (*103*); Jung et al. (2020)(*57*)
AI	aridity index		30 arc-seconds	Global Aridity Index and Potential Evapotranspiration Database v3	Zomer et al. (2022)(*59*)
SWC	soil water content	m^3 m^{-3}	0.25°	ERA5 reanalysis	Hersbach et al. (2020)(*58*)